# AIMO Proof Pilot — Final Grade

**Problem (#_ID):** 3_SINDEX **Submission:** model_3

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**Scoring scale (0/1/6/7) — two binary decisions:**

**Step 1. Is the solution essentially complete?**

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (`docs/markschemes.pdf`).

## 1. Final mark

**Final mark** (choose one): $\boxed{0}$

## 2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. `ME1`) where possible.

The solution exhibits a candidate multiset of index 3 and then simply asserts that index 4 is impossible, that $M = 6$, and that no fewer than 12 elements are possible — all without proof. The claimed maximum index $N = 3$ is wrong (the correct value is 5), and even the index-3 construction is not correctly verified (the displayed partition does not partition the stated multiset). With no valid proof that the index is $\leq 6$, there is nothing to credit and the submission does not reach the partial-progress band. Scores **0**.